

## **Deleted File Recovery and Analysis using Autopsy**

| <b>No.</b> | <b>Topics</b>           |
|------------|-------------------------|
| 1.         | Introduction            |
| 2.         | Objective               |
| 3.         | Tools Used              |
| 4.         | Scope of Project        |
| 5.         | Limitation              |
| 6.         | Autopsy Installation    |
| 7.         | Overview of Tool        |
| 8.         | Practical Demonstration |
| 9.         | Analysis                |
| 10.        | Findings                |
| 11.        | Conclusion              |
| 12.        | References              |

## 1. Introduction:

- Digital Forensics (DFIR) is used to collect and analyze digital data for investigation. It helps to find evidence from computers and devices.
- Tools like **Autopsy** are used to recover deleted files and study user activity.

## 2. Objective:

- The main purpose of this project is to perform deleted file recovery and analyze digital evidence using forensic techniques.
- The goals of this project are:
  1. To recover deleted files from a storage device
  2. To analyze file metadata and timestamps
  3. To understand user activity through recovered data
  4. To gain practical knowledge of digital forensic tools

## 3. Tools Used:

- **Autopsy** – Used for recovering deleted files and analysing digital evidence.

## 4. Scope of Project:

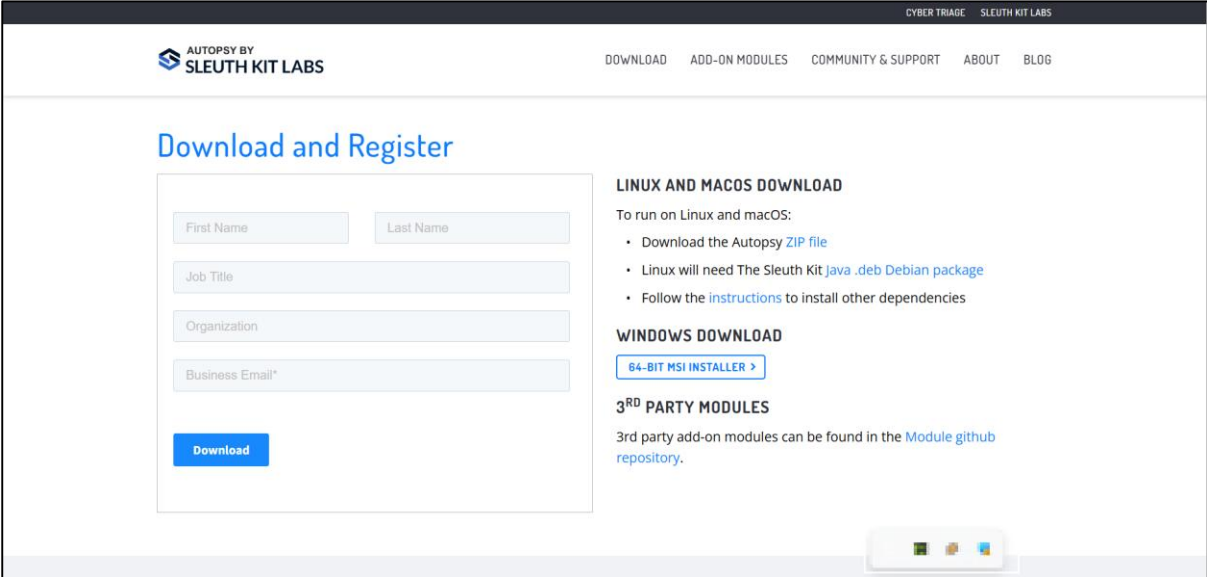
- This project focuses on recovering deleted files and analysing them using Autopsy. It includes file recovery, metadata analysis, and timeline study.

## 5. Limitations:

- This project does not cover advanced forensic techniques like memory analysis or network forensics.

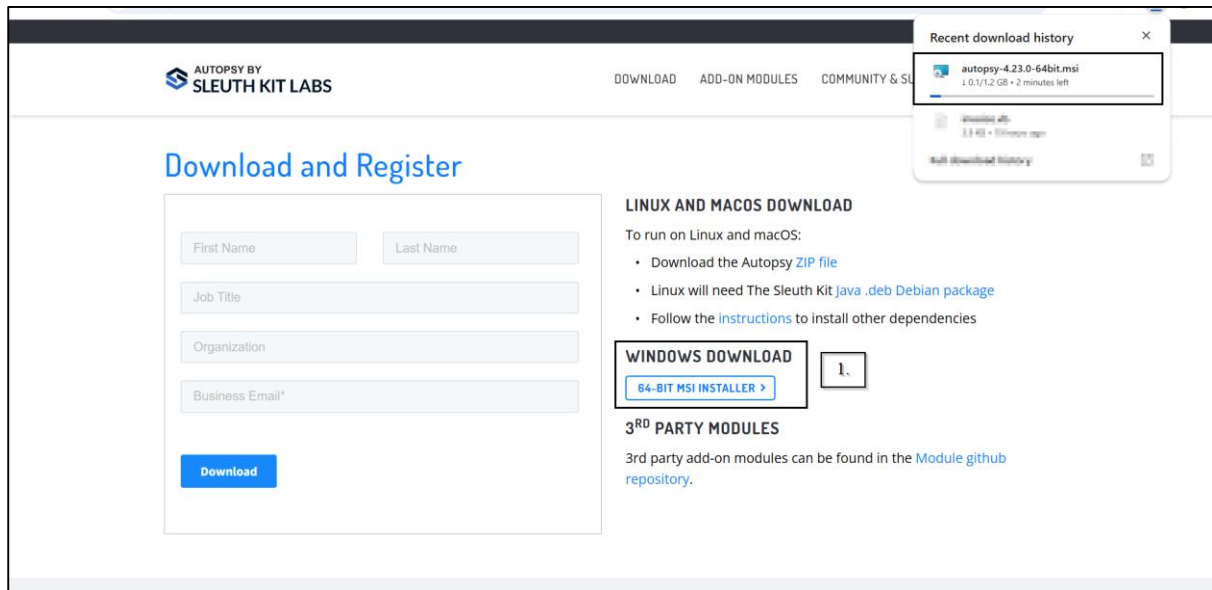
## 6. Autopsy Installation:

**Step-1:** Open browser and then search the “[Autopsy Download](#)” and the interface is displayed.

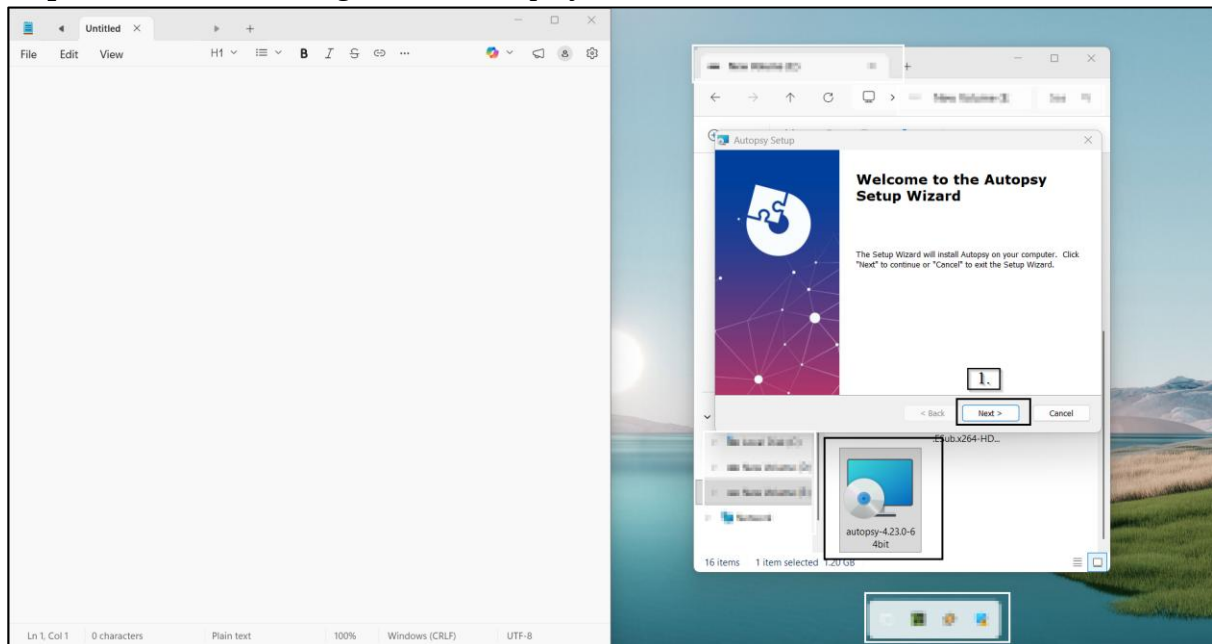


The screenshot shows the Autopsy by Sleuth Kit Labs website. The header includes the logo and navigation links: DOWNLOAD, ADD-ON MODULES, COMMUNITY & SUPPORT, ABOUT, and BLOG. The main heading is "Download and Register". On the left, there is a registration form with fields for First Name, Last Name, Job Title, Organization, and Business Email\*, followed by a blue "Download" button. On the right, there are sections for "LINUX AND MACOS DOWNLOAD" and "WINDOWS DOWNLOAD". The Linux/MacOS section provides instructions and links for downloading the ZIP file, the .deb package, and following installation instructions. The Windows section features a button for the "64-BIT MSI INSTALLER >". Below these, there is a section for "3RD PARTY MODULES" with a link to the Module github repository. The footer shows a Windows taskbar with icons for a file explorer, a terminal, and a web browser.

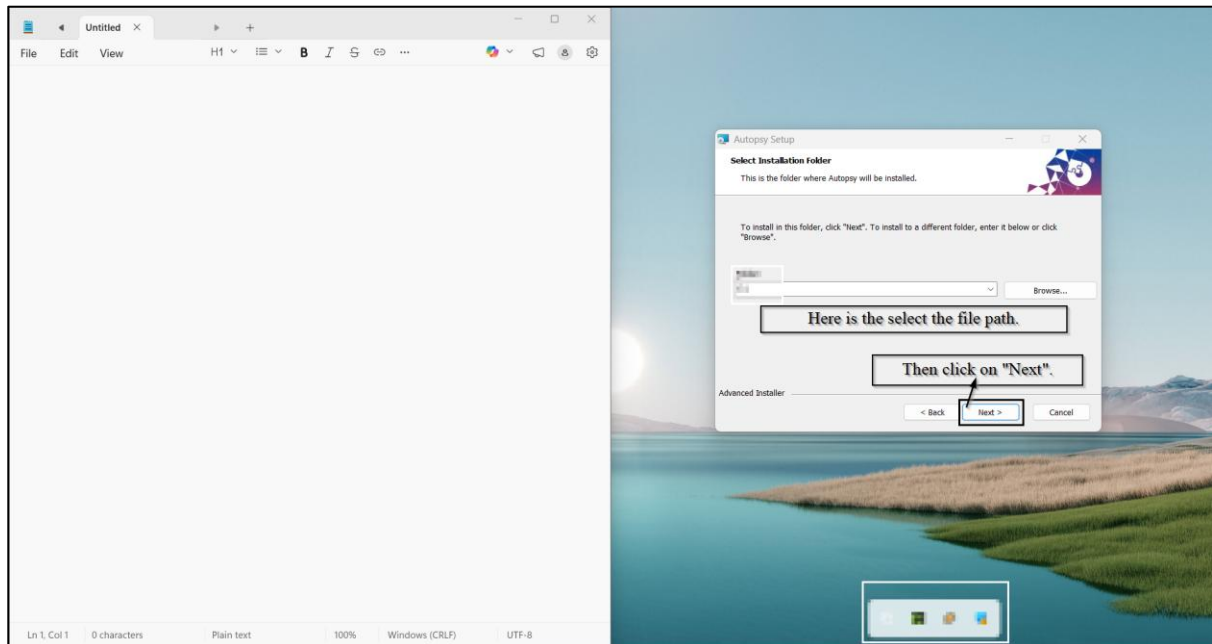
**Step-2:** After opening this link and then click on marked place and download this tool via link.



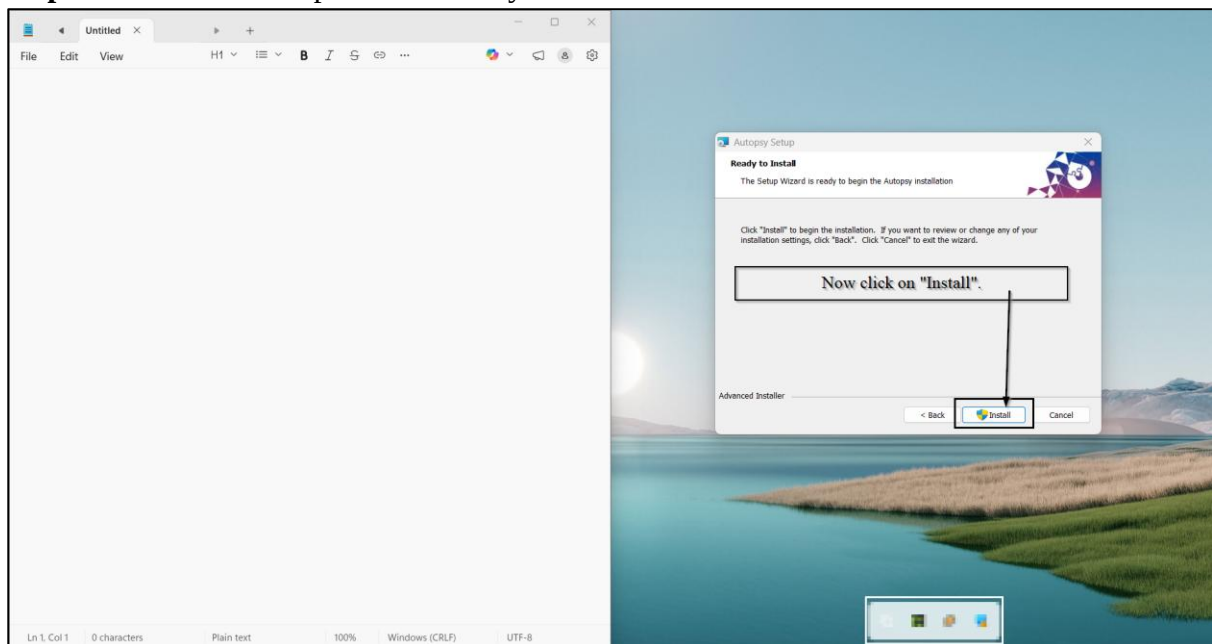
**Step-3:** After downloading, the file is displayed. Now click on this file and then click on **Next**.



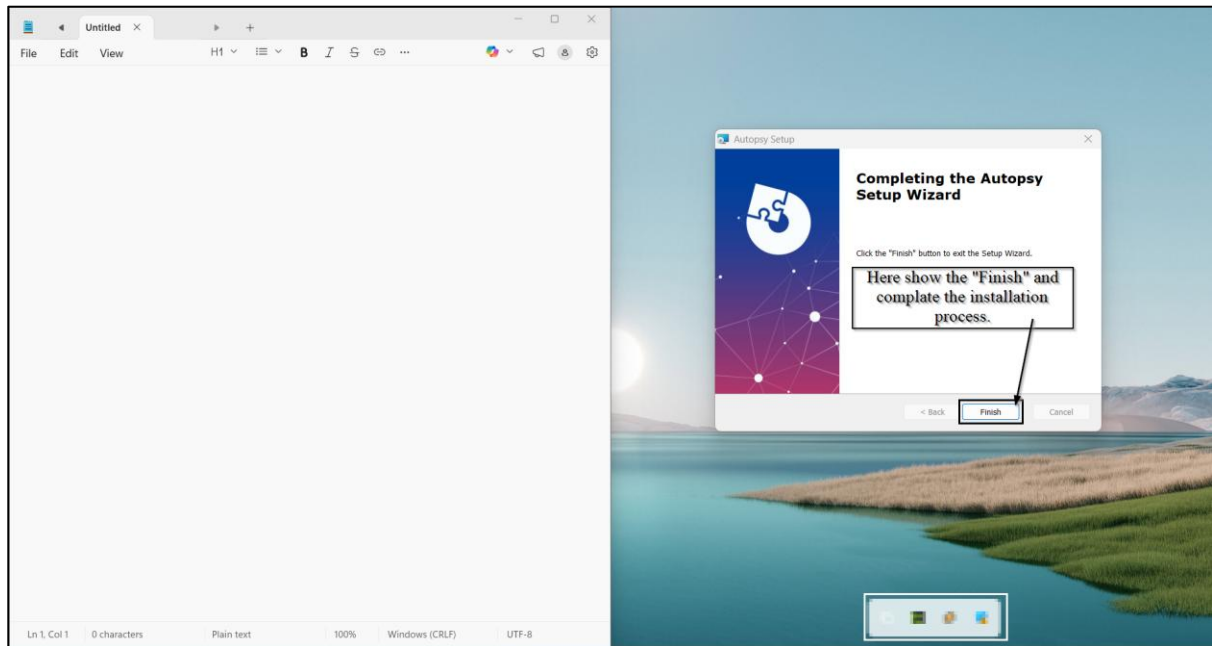
**Step-4:** After then clicking next, select the path where the file will be stored and then one more time click on **Next**.



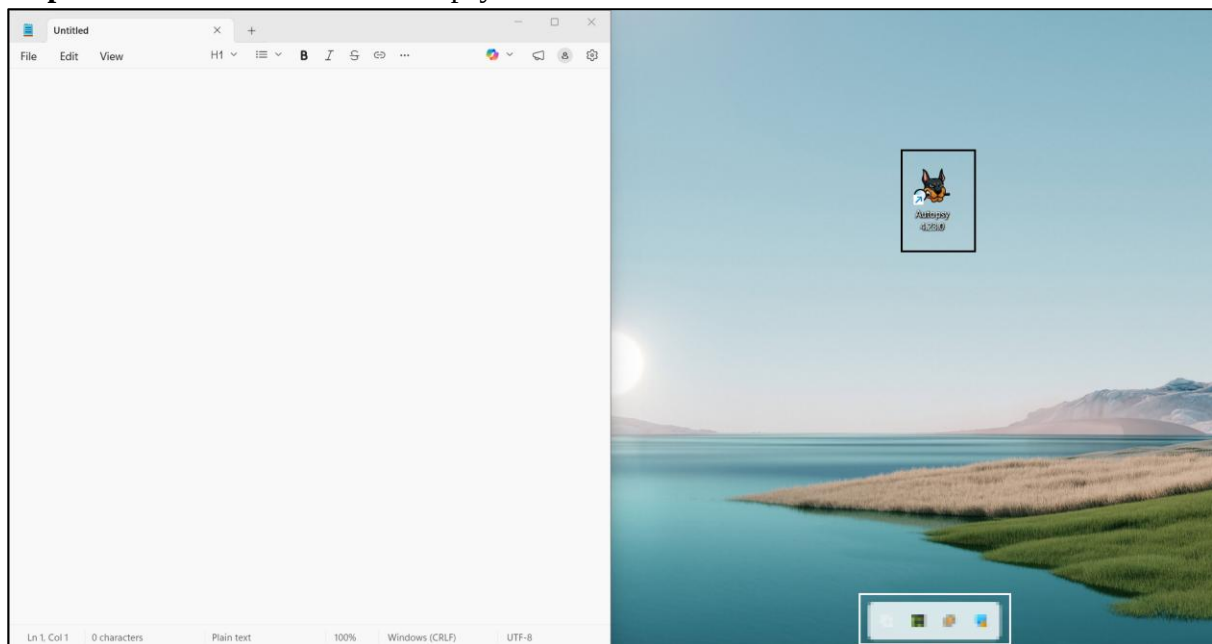
**Step-5:** After select the path then finally show the **Install** button and click on this button.



**Step-6:** After installation then show the **Finish** button. That means the Autopsy tool installed successfully.

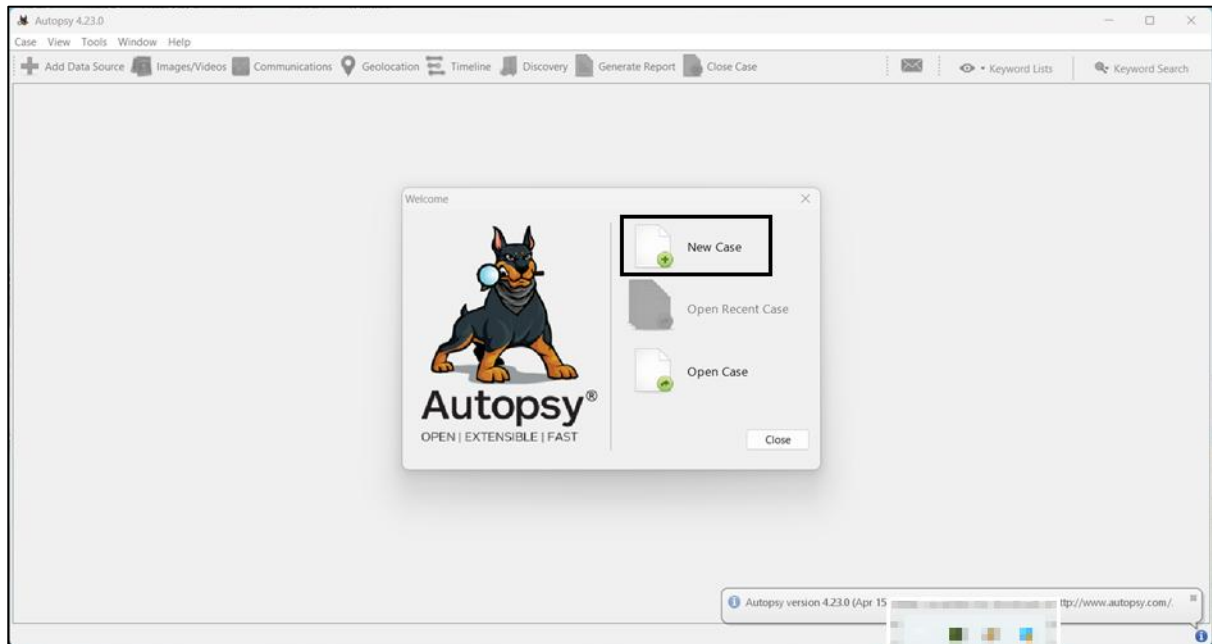


**Step-7:** Here it is the **Icon** of Autopsy tool.

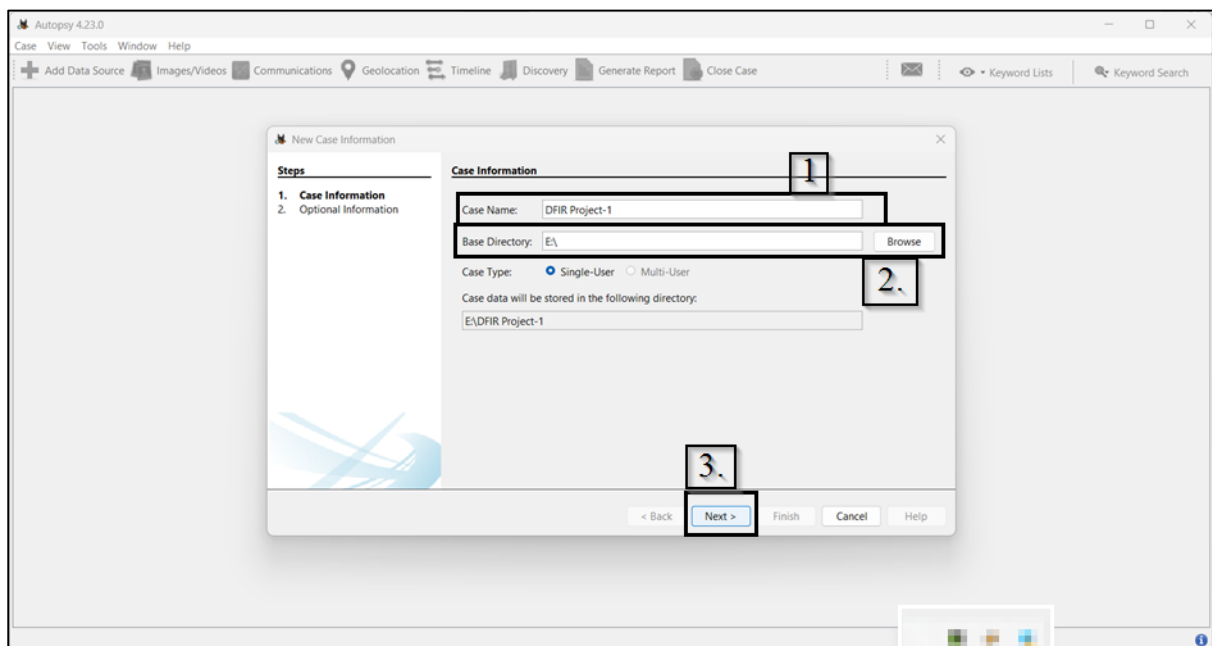


## 7. Overview of Tool /How to setup tool for use:

**Step-1:** Here the first interface of the Autopsy tool. Now click on **New Case**.



**Step-2:** After click on create case then show type of interface. Here first we can write the **Case Name** and then select the **Base Directory** which is store the data of this case and then click on **Next**.



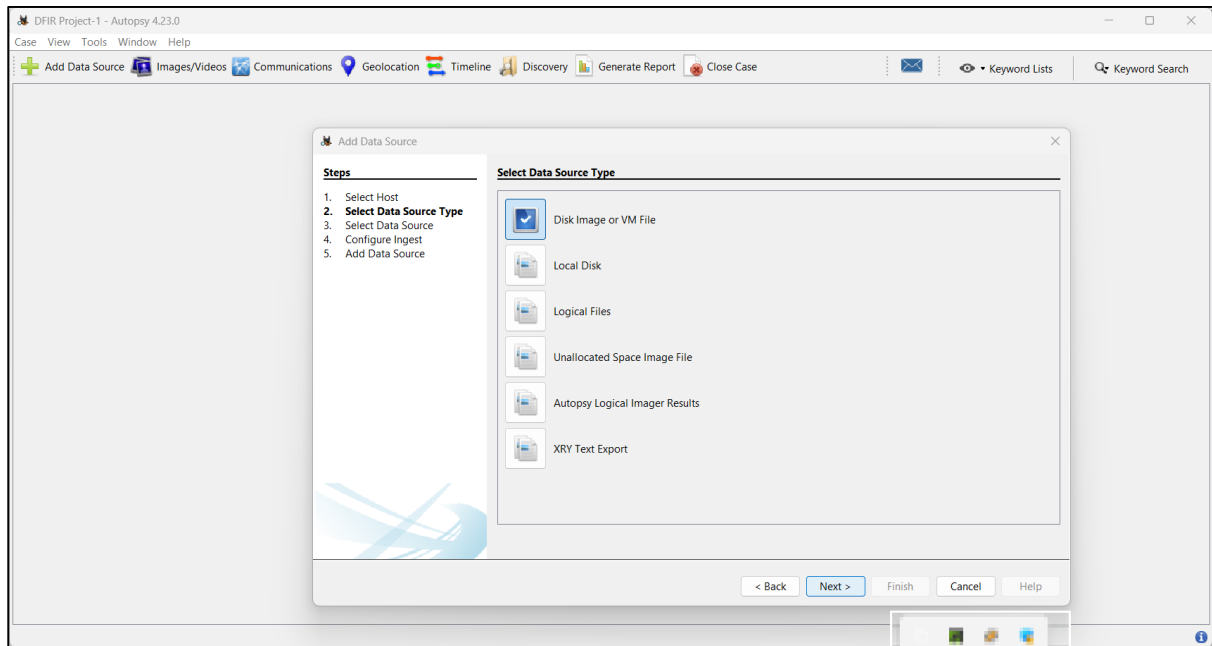
**Step-3:** After then show this type of interface. So here it is optional so direct click on **Finish**.

The screenshot shows the 'New Case Information' dialog box in Autopsy 4.23.0. The 'Optional Information' tab is selected, displaying fields for Case Number, Examiner Name, Phone, Email, Notes, and Organization. The 'Finish' button at the bottom right is highlighted with a red box.

**Step-4:** It is also optional so direct click on **Next**.

The screenshot shows the 'Add Data Source' dialog box in Autopsy 4.23.0. The 'Select Host' tab is selected, displaying options to generate a new host name or use an existing host. The 'Next >' button at the bottom right is highlighted with a red box.

**Step-5:** Now we select the which devices connect to this tool.



**Disk Image / VM File:**

- Used to analyze a full disk copy
- **Devices:** HDD, USB, Virtual Machine

**Local Disk:**

- Used to analyze the real system
- **Devices:** C: drive, D: drive,

**Logical Files:**

- Used to analyze selected files or folders
- **Devices:** Folder, USB, SSD, Hard Disk

**Unallocated Space Image File:**

- Used to analyze deleted data only
- **Device:** Disk free space

**Autopsy Logical Imager Results:**

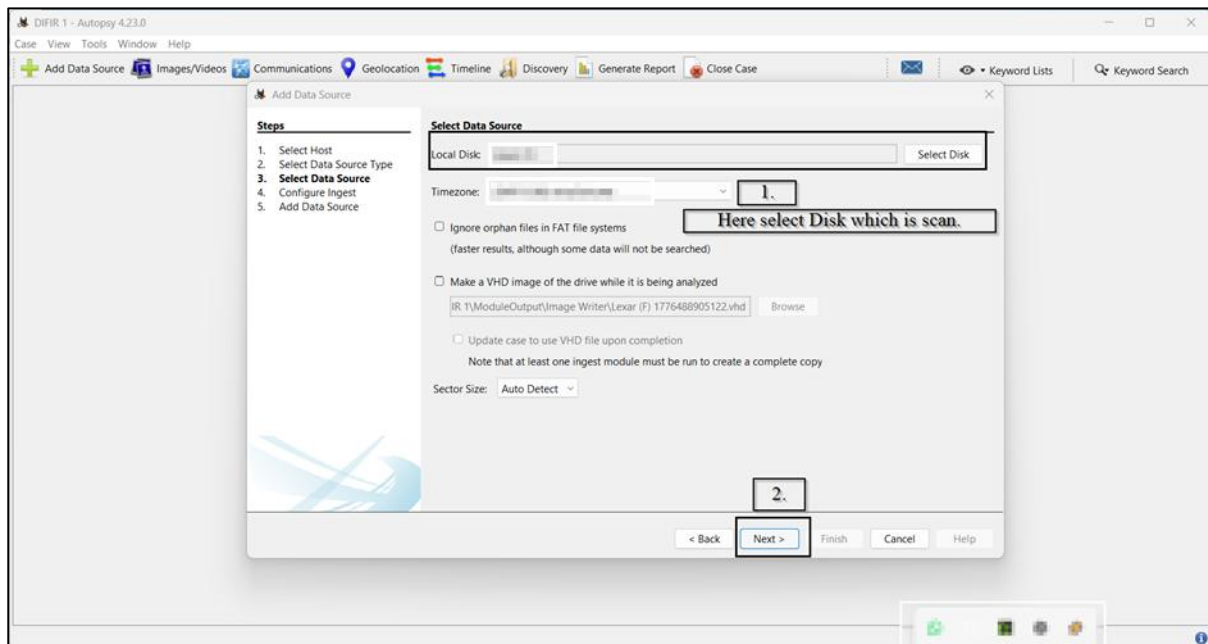
- Used to load data collected by Autopsy
- **Device:** Collected forensic data

**XRY Text Export:**

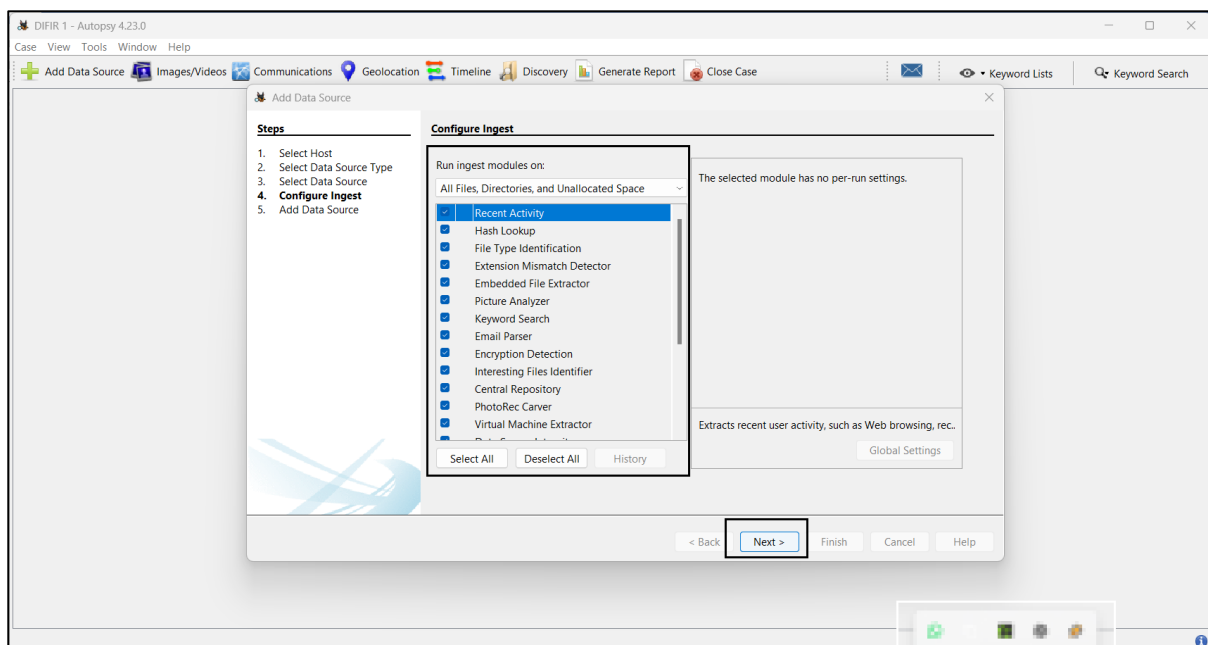
- Used to analyze mobile data
- **Device:** Mobile phone



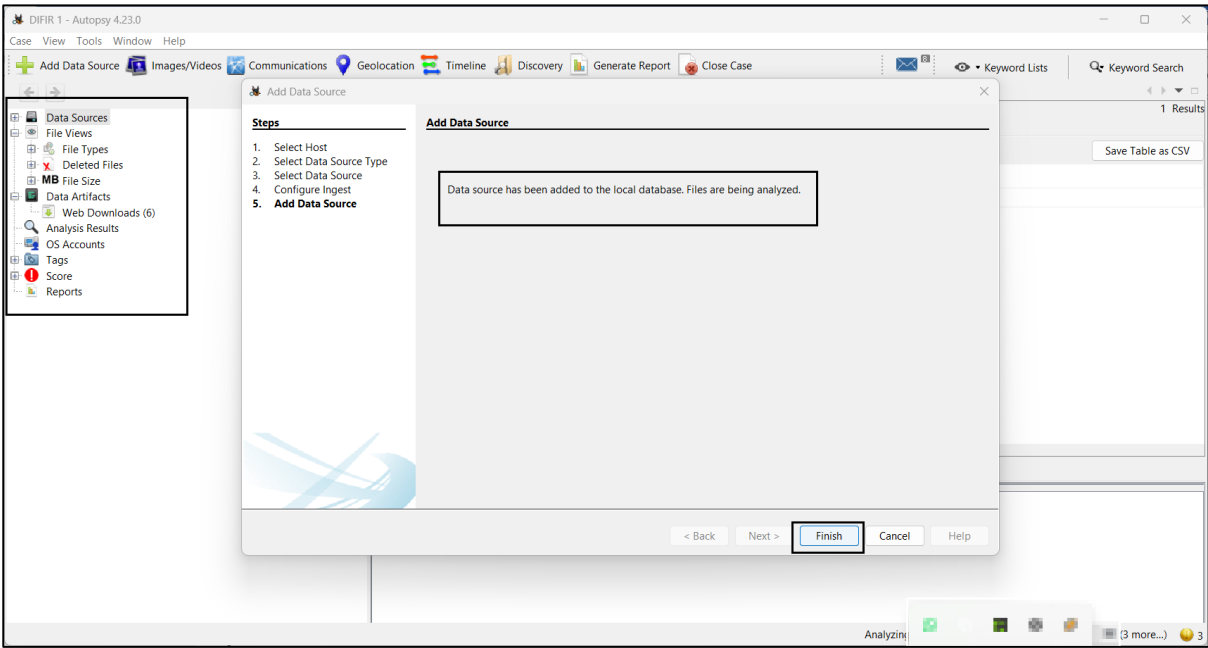
**Step-6:** Select the Data Source Type and then choose the Disk you want to analyze. In this case, a USB drive (F:) is selected as the data source. Click on Select Disk, choose the required drive, and then click Next to continue.



**Step-7:** In this step, select the required options to analyze the data. Choose the necessary modules based on your investigation needs. After selecting the options, click on Next to continue.



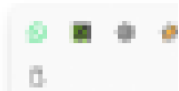
**Step-8:** After selecting the options, the system takes some time to process the data. Once the process is completed, a new interface is displayed, which indicates that the data source has been added successfully. The added data source can be seen in the background.



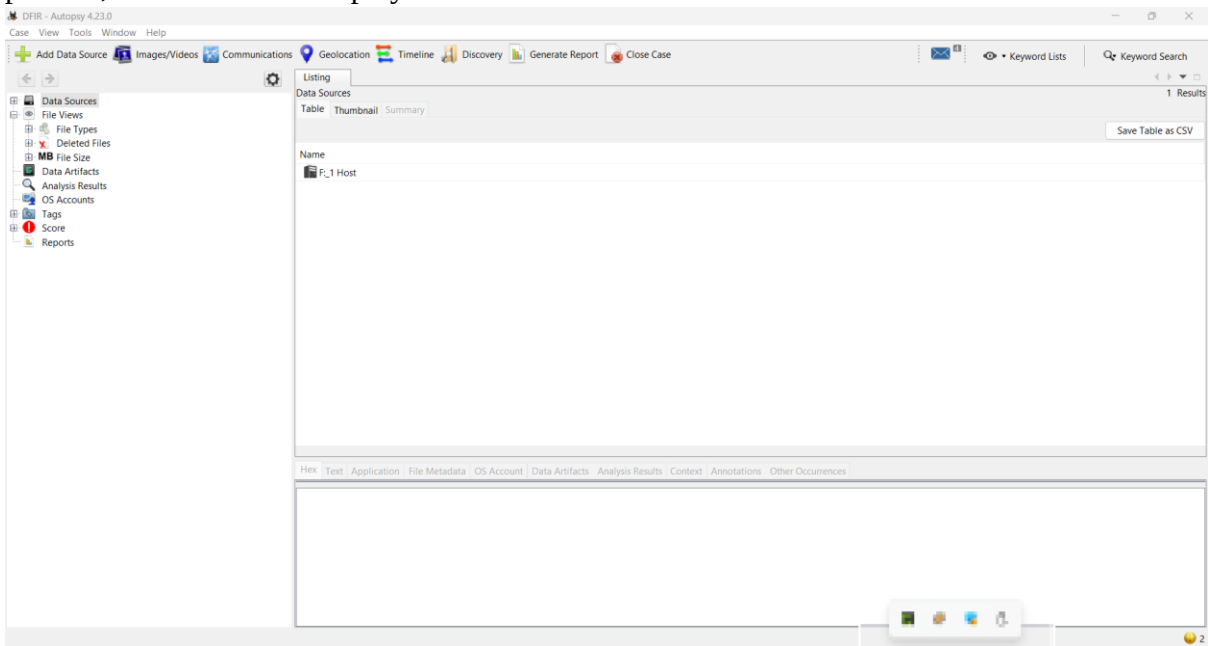
## 8. Practical Demonstration:

- Here I am using this file for the Autopsy tool. This file list is only for learning purpose. The files are attached using a USB device.

| Name                                       | Date modified    | Type                 | Size     |
|--------------------------------------------|------------------|----------------------|----------|
| abcd                                       | 07-04-2026 16:48 | APK File             | 2,278 KB |
| DIFIR Sample 2                             | 23-12-2025 22:56 | Microsoft Edge PD... | 3,621 KB |
| DIFIR Project                              | 18-04-2026 13:23 | Microsoft Edge PD... | 4 KB     |
| ISCON                                      | 09-04-2026 23:44 | JPEG File            | 123 KB   |
| Kaaya AI Try-On Awareness Video (1)        | 17-07-2025 07:34 | MP4 File             | 3,946 KB |
| LEVELS - Official Video Sidhu Moose Wal... | 18-04-2026 10:56 | MP3 File             | 3,658 KB |
| Lifestyle (feat. Banka) [bnutriVrQGg]      | 18-04-2026 10:58 | MP3 File             | 3,962 KB |
| Sample 1                                   | 09-04-2026 14:41 | WEBP File            | 111 KB   |
| Sample 2                                   | 09-04-2026 14:41 | WEBP File            | 78 KB    |
| Sample 4                                   | 09-04-2026 14:41 | WEBP File            | 45 KB    |
| test.txt                                   | 09-11-2025 18:27 | Text Document        | 1 KB     |



**Step-1:** Open the **Autopsy** tool. Create case and then select the devices. After completing the process, the interface is displayed.



**Step-2:** Here in my External devices, there are many items like this.

| Name                                       | Date modified    | Type                 | Size     |
|--------------------------------------------|------------------|----------------------|----------|
| abcd                                       | 07-04-2026 16:48 | APK File             | 2,278 KB |
| DFIR Sample 2                              | 23-12-2025 22:56 | Microsoft Edge PD... | 3,621 KB |
| DIFR Project                               | 18-04-2026 13:23 | Microsoft Edge PD... | 4 KB     |
| ISCON                                      | 09-04-2026 23:44 | JPEG File            | 123 KB   |
| Kaaya AI Try-On Awareness Video (1)        | 17-07-2025 07:34 | MP4 File             | 3,946 KB |
| LEVELS - Official Video Sidhu Moose Wal... | 18-04-2026 10:56 | MP3 File             | 3,658 KB |
| Lifestyle (feat. Banka) [bnutriVrQGs]      | 18-04-2026 10:58 | MP3 File             | 3,962 KB |
| Sample 1                                   | 09-04-2026 14:41 | WEBP File            | 111 KB   |
| Sample 2                                   | 09-04-2026 14:41 | WEBP File            | 78 KB    |
| Sample 4                                   | 09-04-2026 14:41 | WEBP File            | 45 KB    |
| test.txt                                   | 09-11-2025 18:27 | Text Document        | 1 KB     |

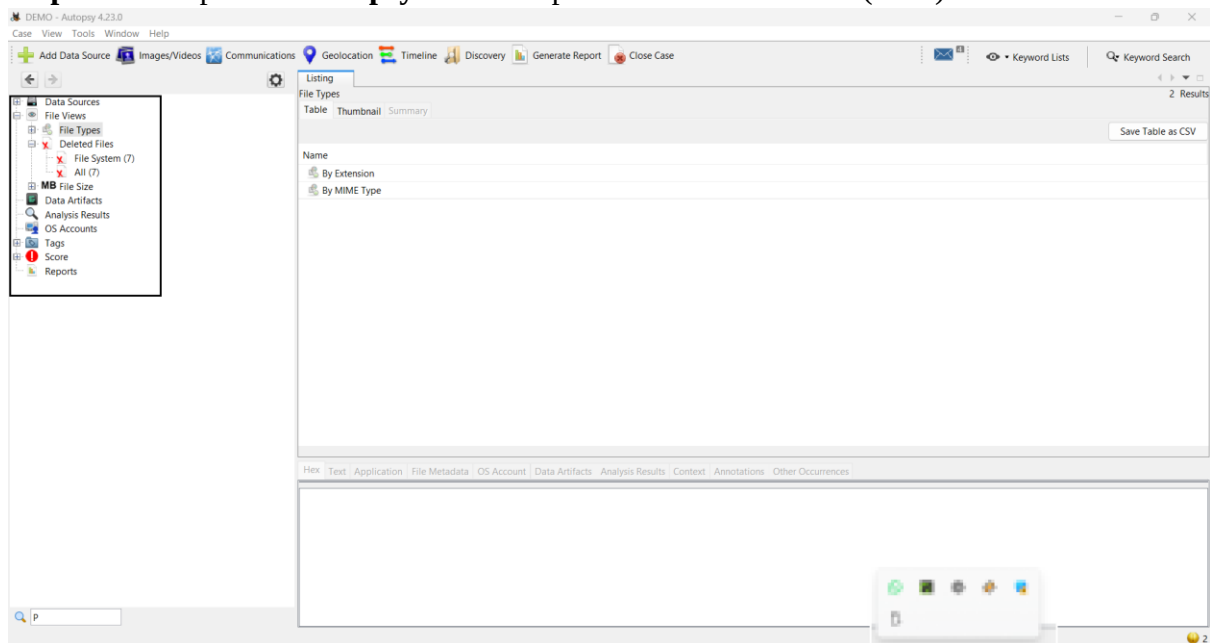


### Step-3: Now here I am deleting some file into this USB.

| Name                                  | Date modified    | Type                 | Size     |
|---------------------------------------|------------------|----------------------|----------|
| DFIR Sample 2                         | 23-12-2025 22:56 | Microsoft Edge PD... | 3,621 KB |
| ISCON                                 | 09-04-2026 23:44 | JPEG File            | 123 KB   |
| Kaaya AI Try-On Awareness Video (1)   | 17-07-2025 07:34 | MP4 File             | 3,946 KB |
| Lifestyle (feat. Banka) [bnutriVrQGs] | 18-04-2026 10:58 | MP3 File             | 3,962 KB |
| Sample 1                              | 09-04-2026 14:41 | WEBP File            | 111 KB   |
| Sample 4                              | 09-04-2026 14:41 | WEBP File            | 45 KB    |
| test.txt                              | 09-11-2025 18:27 | Text Document        | 1 KB     |

Here I am delete one Audio File, one Image,one PDF and one abcd.

### Step-4: Now open the Autopsy tool and open this external device (USB).



- Here, all components and their details are displayed.
- ❖ **Data Sources**
  - This section shows all the devices or data that are added for analysis, such as USB drives, hard disks, or disk images. It is the main source of evidence in the investigation.

## ❖ File Views

### ➤ File Types

- This section displays files grouped by their type, such as images, documents, videos, etc. It helps in quickly finding specific types of files.

### ➤ Deleted Files

- This section shows files that were deleted but are still recoverable. These files are very important in digital forensic investigations as they may contain hidden or important evidence.

### ➤ File Size

- This section organizes files based on their size. It helps investigators identify large or unusual files that may be suspicious.

## ❖ Data Artifacts

- This section shows useful information extracted from the system, such as metadata and web downloads. It helps in understanding user activity and behaviour.

## ❖ Analysis Results

- This section displays results generated by the tool after analysis, such as extension mismatches or suspicious files. It helps in identifying hidden or malicious data.

## ❖ OS Accounts

- This section shows user accounts present on the system. It helps identify who used the system and when.

## ❖ Tags

- This section allows marking important files as evidence. It helps in organizing and highlighting key findings.

## ❖ Score

- This section shows the priority or importance level of files based on analysis. It helps investigators focus on important data.

## ❖ Reports

- This section is used to generate and view the final forensic report. It contains all findings and analysis results.

**Step-5:** After following this process, all deleted content is displayed.

- Here, additional content is added to demonstrate that deleted files can still be recovered and viewed.

The screenshot shows the Autopsy 4.23.0 interface. On the left, the 'Data Sources' pane shows a tree structure with 'File Views' expanded, and 'Deleted Files' selected. A callout box with numbers 1, 2, 3, and 4 points to different parts of this tree. The main pane displays a table of deleted files. A callout box points to a file in the table with the text: 'Click on a deleted file to view its content below. The display changes based on file type (image, PDF, etc.).' The bottom pane shows a preview of a file, which is a dragon fruit image.

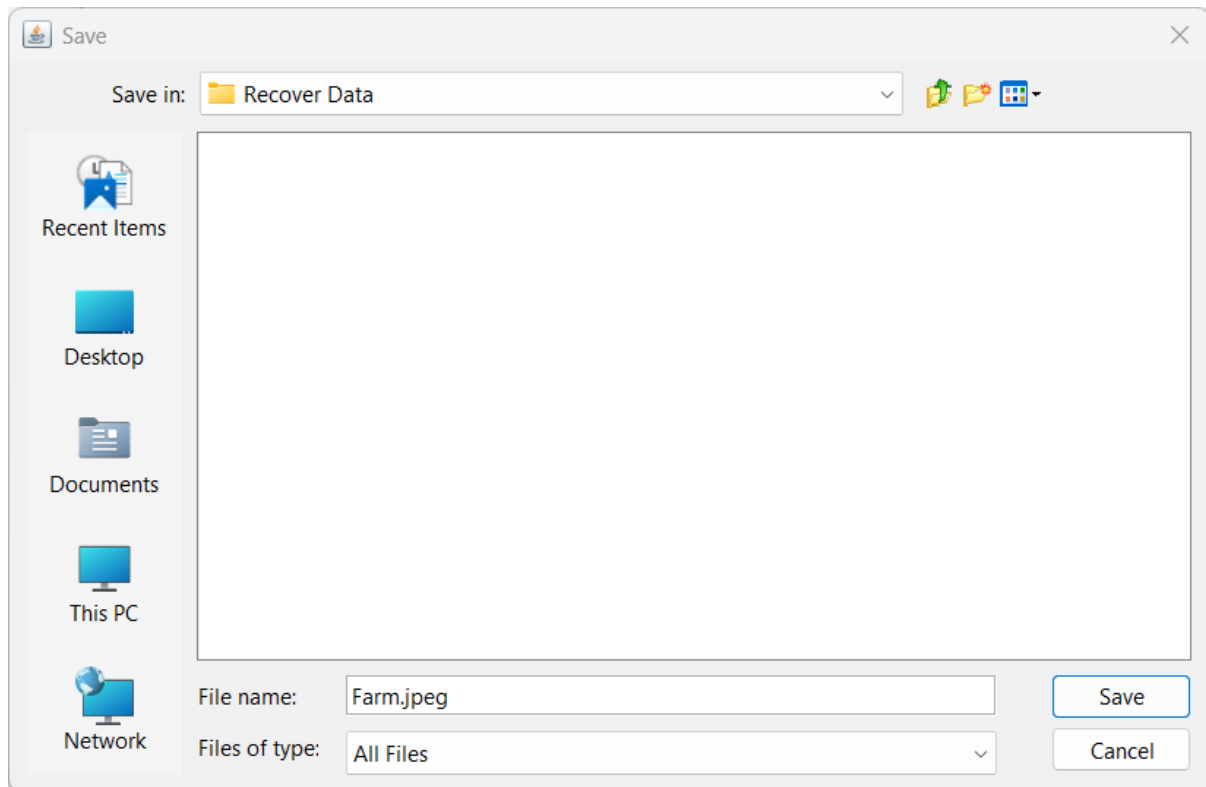
| Name                             | S | C | O | Modified Time           | Change Time             | Access Time             | Created Time            | Size    | Flags(Dir)  | Flags(Meta) |
|----------------------------------|---|---|---|-------------------------|-------------------------|-------------------------|-------------------------|---------|-------------|-------------|
| Farm.jpeg                        |   |   |   | 2026-04-09 23:44:29 IST | 2026-04-13 10:49:30 IST | 2026-04-24 21:38:02 IST | 2026-04-24 21:38:02 IST | 79857   | Unallocated | Unallocated |
| Farm.jpegZone.Identifier         |   |   |   | 2026-04-09 23:44:29 IST | 2026-04-13 10:49:30 IST | 2026-04-24 21:38:02 IST | 2026-04-24 21:38:02 IST | 60      | Unallocated | Unallocated |
| Sample 2.webp                    |   |   |   | 2026-04-09 14:41:38 IST | 2026-04-18 10:54:20 IST | 2026-04-24 21:04:05 IST | 2026-04-15 21:21:04 IST | 78987   | Unallocated | Unallocated |
| Sample 2.webpZone.Identifier     |   |   |   | 2026-04-09 14:41:38 IST | 2026-04-18 10:54:20 IST | 2026-04-24 21:04:05 IST | 2026-04-15 21:21:04 IST | 60      | Unallocated | Unallocated |
| DIFR Project.pdf                 |   |   |   | 2026-04-18 13:23:26 IST | 2026-04-18 13:23:26 IST | 2026-04-18 13:23:55 IST | 2026-04-18 13:23:55 IST | 3197    | Unallocated | Unallocated |
| Dragon Fruit.jpeg                |   |   |   | 2026-04-09 23:45:37 IST | 2026-04-13 10:49:30 IST | 2026-04-24 21:38:02 IST | 2026-04-24 21:38:02 IST | 130732  | Unallocated | Unallocated |
| Dragon Fruit.jpegZone.Identifier |   |   |   | 2026-04-09 23:45:37 IST | 2026-04-13 10:49:30 IST | 2026-04-24 21:38:02 IST | 2026-04-24 21:38:02 IST | 60      | Unallocated | Unallocated |
| abcd.apk                         |   |   |   | 2026-04-07 16:48:48 IST | 2026-04-07 16:48:53 IST | 2026-04-16 10:11:37 IST | 2026-04-16 10:11:36 IST | 2332330 | Unallocated | Unallocated |
| 2026-01-07 16:20:35 IST          |   |   |   | 2026-01-07 16:21:33 IST | 2026-04-15 21:22:53 IST | 2026-04-15 21:22:53 IST | 2026-04-15 21:22:53 IST | 3696747 | Unallocated | Unallocated |

**Step-6:** Now, the image and other files are recovered. First, click on the file you want to recover.

The screenshot shows the Autopsy 4.23.0 interface. On the left, the 'Data Sources' pane shows a tree structure with 'File Views' expanded, and 'Deleted Files' selected. A callout box with numbers 1 and 2 points to different parts of this tree. The main pane displays a table of deleted files. A callout box points to a file in the table with the text: 'Right Click on Here and show this type of List like Pop-up.' The bottom pane shows a preview of a file, which is a sunset image.

| Name                             | S | C | O | Modified Time           | Change Time             | Access Time             | Created Time            | Size    | Flags(Dir)  | Flags(Meta) |
|----------------------------------|---|---|---|-------------------------|-------------------------|-------------------------|-------------------------|---------|-------------|-------------|
| Farm.jpeg                        |   |   |   | 2026-04-09 23:44:29 IST | 2026-04-13 10:49:30 IST | 2026-04-24 21:38:02 IST | 2026-04-24 21:38:02 IST | 79857   | Unallocated | Unallocated |
| Farm.jpegZone.Identifier         |   |   |   | 2026-04-09 23:44:29 IST | 2026-04-13 10:49:30 IST | 2026-04-24 21:38:02 IST | 2026-04-24 21:38:02 IST | 60      | Unallocated | Unallocated |
| Sample 2.webp                    |   |   |   | 2026-04-09 14:41:38 IST | 2026-04-18 10:54:20 IST | 2026-04-24 21:04:05 IST | 2026-04-15 21:21:04 IST | 78987   | Unallocated | Unallocated |
| Sample 2.webpZone.Identifier     |   |   |   | 2026-04-09 14:41:38 IST | 2026-04-18 10:54:20 IST | 2026-04-24 21:04:05 IST | 2026-04-15 21:21:04 IST | 60      | Unallocated | Unallocated |
| DIFR Project.pdf                 |   |   |   | 2026-04-18 13:23:26 IST | 2026-04-18 13:23:26 IST | 2026-04-18 13:23:55 IST | 2026-04-18 13:23:55 IST | 3197    | Unallocated | Unallocated |
| Dragon Fruit.jpeg                |   |   |   | 2026-04-09 23:45:37 IST | 2026-04-13 10:49:30 IST | 2026-04-24 21:38:02 IST | 2026-04-24 21:38:02 IST | 130732  | Unallocated | Unallocated |
| Dragon Fruit.jpegZone.Identifier |   |   |   | 2026-04-09 23:45:37 IST | 2026-04-13 10:49:30 IST | 2026-04-24 21:38:02 IST | 2026-04-24 21:38:02 IST | 60      | Unallocated | Unallocated |
| abcd.apk                         |   |   |   | 2026-04-07 16:48:48 IST | 2026-04-07 16:48:53 IST | 2026-04-16 10:11:37 IST | 2026-04-16 10:11:36 IST | 2332330 | Unallocated | Unallocated |
| 2026-01-07 16:20:35 IST          |   |   |   | 2026-01-07 16:21:33 IST | 2026-04-15 21:22:53 IST | 2026-04-15 21:22:53 IST | 2026-04-15 21:22:53 IST | 3696747 | Unallocated | Unallocated |

**Step-7:** After click on **Extract File** then show this type of interface. Now select where save this recover data and then click on **Save**.



**Step-8:** Here, I follow the same process to extract the content. In this step, I am extracting data stored on an external device. This process helps to recover data from external devices.



**Analysis:** The deleted files were successfully recovered from the USB device. The recovered files include images and documents. File details such as name, type, size, and date were checked. The timestamps help to understand when the file was created or modified. This analysis shows that deleted data can still be accessed and examined using forensic tools.

**Findings:** Deleted files can be recovered easily. Data is not permanently deleted from the device. File details and timestamps help in investigation. Forensic tools help to understand user activity.

**Conclusion:** Autopsy is a useful tool for recovering deleted files and analysing digital data. This project shows that deleted data can still be recovered and used for investigation. It also helps in understanding basic digital forensic techniques.

## **References**

- Autopsy Official Website: <https://www.autopsy.com/download/>